Nicholas Steffen

He/Him • Warwick, RI • +1 (516) 456-1214 • nicksteffen94@gmail.com • linkedin.com/in/nicholassteffen1 • https://nicksteffen.github.io/portfolio/

Senior Software Engineer

Senior Software Engineer with 10+ years of experience designing and building scalable backend platforms, cloud-native applications, API integrations, and developer tooling across SaaS, defense, and consulting environments. Strong background in Python, Java, distributed systems, cloud infrastructure, and data-intensive applications. Proven ability to modernize legacy systems, improve developer productivity, and deliver reliable, high-performance solutions supporting mission-critical operations. Combines hands-on engineering expertise with a foundation in Applied Mathematics and strong cross-functional leadership.

WORK EXPERIENCE

Independent Consultant • Remote • 12/2025 - Present

Lead Full-Stack Consultant

- Designed and delivered a scalable, serverless backend using Supabase Edge Functions and PostgreSQL, migrating from a FastAPI prototype to optimize for performance and cost.
- Built and maintained a custom CI/CD pipeline on Vercel with multi-branch environments (Main, Develop, Staging, Production) to safely vet AI-assisted code contributions and improve release reliability.
- Engineered core platform features including secure authentication and cross-platform payment integration, supporting real-time synchronization across web and mobile clients.
- Directed technical launch operations, establishing health monitoring and analytics to track performance during high-traffic events.
- Collaborated with mobile engineering teams to align database schema designs and backend infrastructure with iOS-specific constraints.
- Currently architecting a commission sales reporting tool for early-stage sales operations, focusing on backend data pipelines and reporting APIs.

ClearlyRated • Remote • 06/2021 - 11/2024

Software Engineer

- Served as technical owner of a reusable, framework-agnostic Java object model for third-party API integrations, accelerating CRM onboarding (Salesforce, Microsoft Dynamics) and enabling rapid UI development against diverse data sources.
- Led backend modernization from Flask to Quart for async capabilities, resolving long-standing logging issues and configuring Hypercorn for robust production deployment with minimal disruption.
- Developed internal developer tooling to containerize MySQL databases using Docker and Python scripts that generated reproducible, production-scale datasets, reducing environment setup time and improving testing fidelity for CI/CD pipelines.
- Led research to enhance unit-test efficiency, identifying CI/CD bottlenecks and implementing JUnit-based refactoring with mocks to accelerate test cycles and developer productivity.
- Engineered full-stack solutions (Vue.js, Java, MySQL) for CSV contact data ingestion, validation, and processing, abstracting backend logic to expand integration capabilities.
- Partnered with cross-functional teams on API architecture, system integration, and internal platform improvements supporting developer productivity.

Raytheon • Portsmouth, RI • 08/2018 - 05/2021

Software Engineer II

- Architected and deployed a Python-based device monitoring platform supporting mission-critical hardware operations, integrating legacy Java services through Redis and automating preventative shutdown workflows to reduce equipment failure risk and improve operational reliability.
- Led research and implementation of observability infrastructure for defense hardware systems with 99% uptime requirements, evaluating and deploying Splunk with dual-instance failover architecture to ensure continuous data capture.
- Developed Python API scripts for redundant data transmission to monitoring instances, minimizing data loss across distributed hardware environments.
- Authored Ansible playbooks and Kickstart files for automated provisioning and configuration management across fleets of hardware devices and monitoring servers, streamlining deployment and maintaining consistency.
- Presented architectural trade-offs and tooling recommendations to stakeholders, influencing platform decisions for mission-critical monitoring systems.

Computer Associates Inc. • Smithfield, RI • 06/2016 - 07/2018

Software Consultant

- Collaborated with enterprise clients to gather requirements and customize core ERP software, delivering client-specific web applications including product configurators and sales data portals.
- Enhanced client self-service capabilities by enabling online product design, quote generation, and sales data access, expanding business reach and improving customer engagement.

Farsounder • Warwick, RI • 05/2015 - 05/2016

Software Engineer Intern

- Improved C++ map overlays for real-time GPS tracking and navigation, applying trigonometry to calculate distance and angle from ship heading to user-defined points.
- Enhanced internal data playback tools with tag support and hash-based dataset lookup, streamlining developer workflows and accelerating local testing of new features.
- Researched feasibility of forward-looking sonar for depth approximation, designing validation methodologies against existing down-looking depth finders.

EDUCATION

Bachelor of Science in Applied Mathematics

Providence College • 09/2012 - 05/2016

Minor: Computer Science

SKILLS

Languages: Python, Java, C++, TypeScript, JavaScript, SQL, Bash

Frameworks & APIs: FastAPI, Flask, Quart, Node.js, React, Next.js, Vue.js, REST APIs, OAuth, JWT

Cloud & Infrastructure: GCP, Docker, Linux/Unix, Ansible, CI/CD, Vercel

Data & Messaging: PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch, RabbitMQ, SQLite

Architecture: Distributed Systems, API Design, Microservices, System Integration, Event-Driven Architecture, Scalability Planning

AI & Developer Tooling: LLM-assisted development (Claude Code, Opencode), AI chatbot guardrails, automated testing frameworks

Other Skills: Scalable Platforms, High Availability Systems, Data Processing, Observability, Performance Optimization, Automation, Mission-Critical Environments